

Javier Deng Xu

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EDUCATION

Simon Fraser University

Burnaby, BC

Bachelors of Computer Science - Software Systems (Co-op)

Expected Graduation: September 2027

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Database Systems (MySQL)
- **Awards:** Safe Software Best Modern C++ Award (StormHacks 2025), Starlit Vision - Best Potential Award (DreamHacks 2025), President's Scholarship (UofWaterloo)

EXPERIENCE

Software Developer

February 2026 – Present

SFU Blueprint | React, Node.js, Express, RESTful API

Burnaby, BC

- Architected dynamic web interfaces with **React**, maintaining **100% state synchronization** across complex user workflows by leveraging optimized Hooks (useState, useEffect) for seamless data integration.
- Developed Full Stack applications using **React Hooks** and **Node.js**, achieving a **30% reduction** in response latency by implementing asynchronous **Express middleware** to handle concurrent **API** traffic.

StormForge (Full-Stack)

January 2026 – Present

SFU Surge | Python, OpenCV, MediaPipe, WebSockets, Docker

Burnaby, BC

- Implemented a real-time **Python** hand-tracking tool using **OpenCV** and **MediaPipe** to correct typing habits and integrated Socket.io and **Flask** to achieve **93.5%** accuracy in coordinate analysis and user feedback.
- Optimized real-time video processing pipelines by implementing **WebSockets** in Python, reducing frame latency by **15%** and ensuring consistent **30 FPS tracking** during high-intensity typing sessions.
- Containerized the computer vision backend using **Docker**, ensuring environment parity across development and production saving time on development for all team members.

Software Developer

September 2025 – Present

SFU Robot Soccer Club | C++, Qt

Burnaby, BC

- Developed a **C++/Qt** path-planning component using RVO2 (ORCA), including a coordinator layer for **agent/obstacle** management, timestep simulation, and signal-driven velocity updates.
- Integrated **RVO2/ORCA** collision avoidance into the path-planning stack to **reduce by 53%** inter-robot collisions (ally/enemy).
- Collaborated cross-functionally with **5+ team leads** in weekly design reviews, delivering **15+ feature tickets** on schedule through **Agile workflows** managed via **Jira** and **GitLab**.

PROJECTS

GANPI (StormHacks 2025 Winner) | C++, Google Gemini's API

October 2025

- Designed and implemented a file summarization feature that generates concise, human-readable summaries of file contents, allowing users to quickly understand documents without opening them.
- Debugged and resolved **critical parsing errors** in API response handling, improving summarization accuracy by **40%** and eliminating **edge-case failures** in file processing.
- Refactored prompt engineering logic to handle **multi-format documents**, ensuring **consistent** summary generation across TXT, PDF, and code files with **95%+ reliability**.

EasyChart (Full-Stack) | React, Tailwind, Node.js, Express.js, CI

August 2025

- Optimized end-to-end data flow by integrating a **React/Tailwind CSS** frontend with a **Express** backend, **reducing** chart rendering latency through efficient routing **optimization** and real-time data processing.
- Set up and configured **Continuous Integration** pipeline for the project using **GitHub Actions**, automating the build process of the application ensuring smooth code integration.

SpikeConnect (DreamHacks 2025 Winner) | React Native, Python

March 2025

- Built SpikeConnect, a **mobile-first social network** for volleyball enthusiasts, implementing player matchmaking algorithms, location-based game discovery, and automated tournament bracketing systems.
- Engineered a dynamic community post section with robust engagement features including likes, comments, and sharing capabilities, fostering active user interaction with **React Native** and **Python**.

TECHNICAL SKILLS

Languages & Databases: Python, C/C++, Node.js, Typescript, Java, MySQL, PostgreSQL

Frameworks/libraries: React, React Native, Bootstrap, Tailwind CSS, Vite, Express, Springboot

Developer Tools: VS Code, Cursor, Linux, Git, GitHub, Figma, Bash, GitHub Actions, Docker, Postman

Transferrable skills: Communication, Team Collaboration, Time Management, Documentation